Technical Report

Emily Wu and Starr Mark

2026-04-15

Results

Our Final First Stage Model is as follows:

$logit(P(AtRisk = 1)) = \beta_0 + \beta_1 YearsofEducation + \beta_2 ADD/ADHD + \beta_3 Gender : Male + \beta_4 Offense : Other + \beta_5 Offense : Property + \beta_6 Offense : PublicOrder + \beta_7 Offense : Unknown + \beta_8 Offense : Violent + \beta_9 Children + \beta_{10} TimesArrested + \beta_{11} MentalHealthTreatment + \beta_{12} Alcoholism$

The first stage of analysis employed logistic regression, a binomial family logit link model. The reported coefficients, β, represent the change in log-odds of being classified as "at risk" for every one-unit increase in the predictor variable. These estimates are modeled according to the logit link function:

$logit(p) = \ln(\frac{p}{1-p}) = \beta_0 + \beta_1 x_1 ... \beta_k x_k$

where p is the probability of risk classification

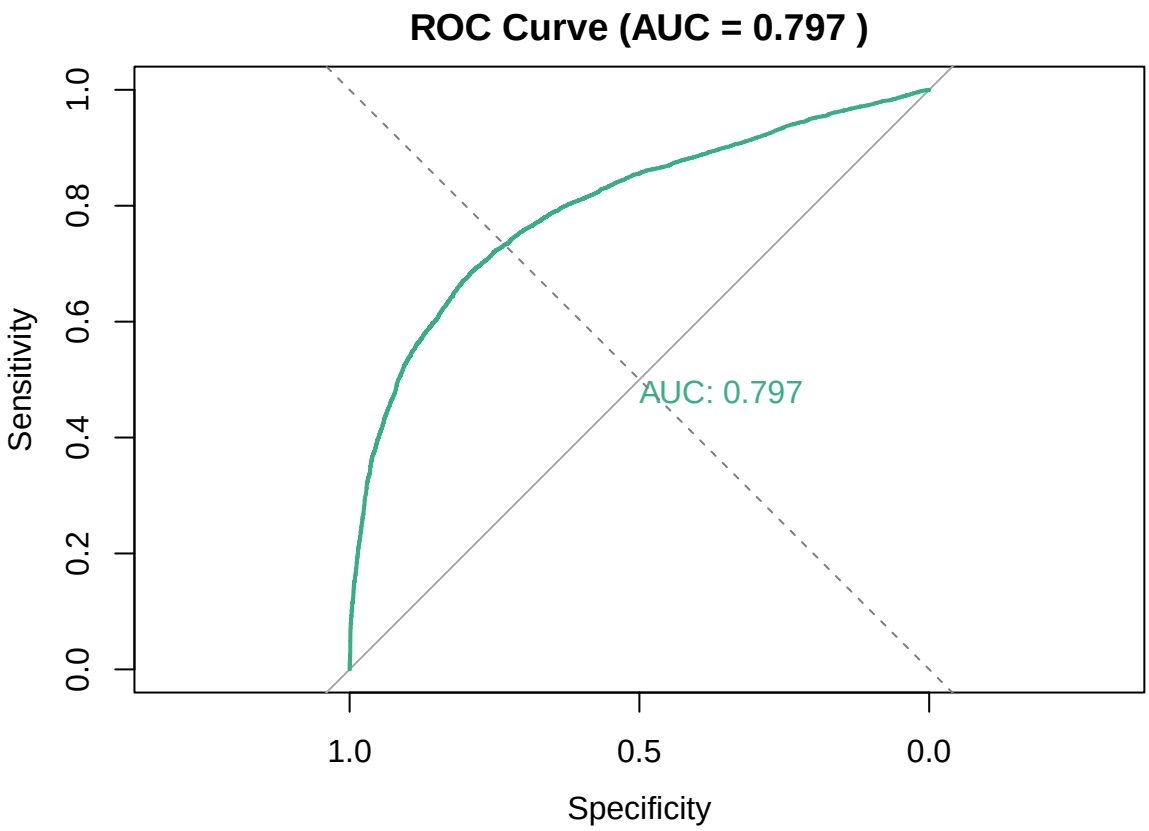
After conducting a Likelihood-Ratio Test on our model, we find that our final model has a deviance score (null vs residual) of 4685 on 12 degrees of freedom and a p-value less than 0.001. This model was fit on $n = 15,450$ observations with 1708 dropped/deleted observations. Over 90% are due to alcoholism.

This indicates our full model significantly outperformed the null model and the combination of clinical and demographic factors provides substantial predictive power for inmate infraction risk. Overall, our model had a 73% classification accuracy (classification of inmates who had infractions or not) and with an AUC of 0.797, it demonstrates a good amount of discriminative ability, correctly distinguishing between at risk and non-at risk individuals about 80% of the time when presented with a random pair of one positive and one negative case.

Table 1: Stage 1 Confusion Matrix

Classification	Observed: No Infraction (0)	Observed: At Risk (1)
Predicted: Low Risk (0)	5,361	2,365
Predicted: High Risk (1)	1,755	5,969

Note:
Overall Accuracy: 73.3% | Sensitivity: 75.3% | Specificity: 71.6% | P-Value: $<$ .001 | No Information Rate (NIR): 53.9%



The model diagnostics state that the most significant predictors are ADD/ADHD, Male Gender, Times Arrested, and Mental Health Treatment. Out of these, the predictors with the largest effect size are ADD/ADHD with 1.33, Male Gender with -0.938, and Mental Health Treatment with 2.201.

Table 2: Significant Predictors

Variable	Estimate	OR	Z_Value	95% Confint IRR
ADD/ADHD	1.3389959	3.8152109	28.22076	(3.48 - 4.19)
Male Gender	-0.9386538	0.3911541	-19.65422	(0.36 - 0.43)
Mental Health Treatment	2.2016663	9.0400643	43.74620	(8.2 - 9.98)

The final Second Stage Model is as follows:

$ln(Infractions) = \beta_0 + \beta_1 Race : Black + \beta_2 Race : Hispanic + \beta_3 Race : Multiracial + \beta_4 Race : Other + \beta_5 Race : White + \beta_6 Number of Diagnoses + \beta_7 Armed Forces + \beta_8 Times Arrested + \beta_9 Years of Education + \beta_{10} ADD/ADHD + \beta_{11} Homeless + \beta_{12} Children + \beta_{13} Mental Health Treatment + \beta_{14} Alcoholism + \beta_{15} Alcohol Treatment + \beta_{16} Gender : Male + \beta_{17} + \beta_{18} Offense : Other + \beta_{19} Offense : Property + \beta_{20} Offense : Public Order + \beta_{21} Offense : Unknown + \beta_{22} Offense : Violent + \beta_{23} Number of Diagnoses * Mental Health Treatment + \beta_{24} Alcoholism * Alcohol Treatment$

The second stage utilized negative binomial regression with a log link to analyze the infraction frequency within the high-risk subgroup. The coefficients are reported on the log-count scale. Due to the model using the natural log link, the relationship is such:

$\ln(\mu) = \beta_0 + \beta_1 x_1 ... \beta_k x_k$

Comparing the 2 models using their goodness-of-fit statistics, it is seen that a negative binomial model is more favorable with a lower AIC and Overdispersion statistic (ϕ). The negative binomial model was fitted on the subpopulation of $n = 7,619$ of those rated as "High Predicted Risk." This was determined by our previous logistic model of those who had a higher predicted probability of having an infraction than the median value of the data set, which was 0.497. 105 observations were not counted, over 90% were deleted due to Race being unavailable.

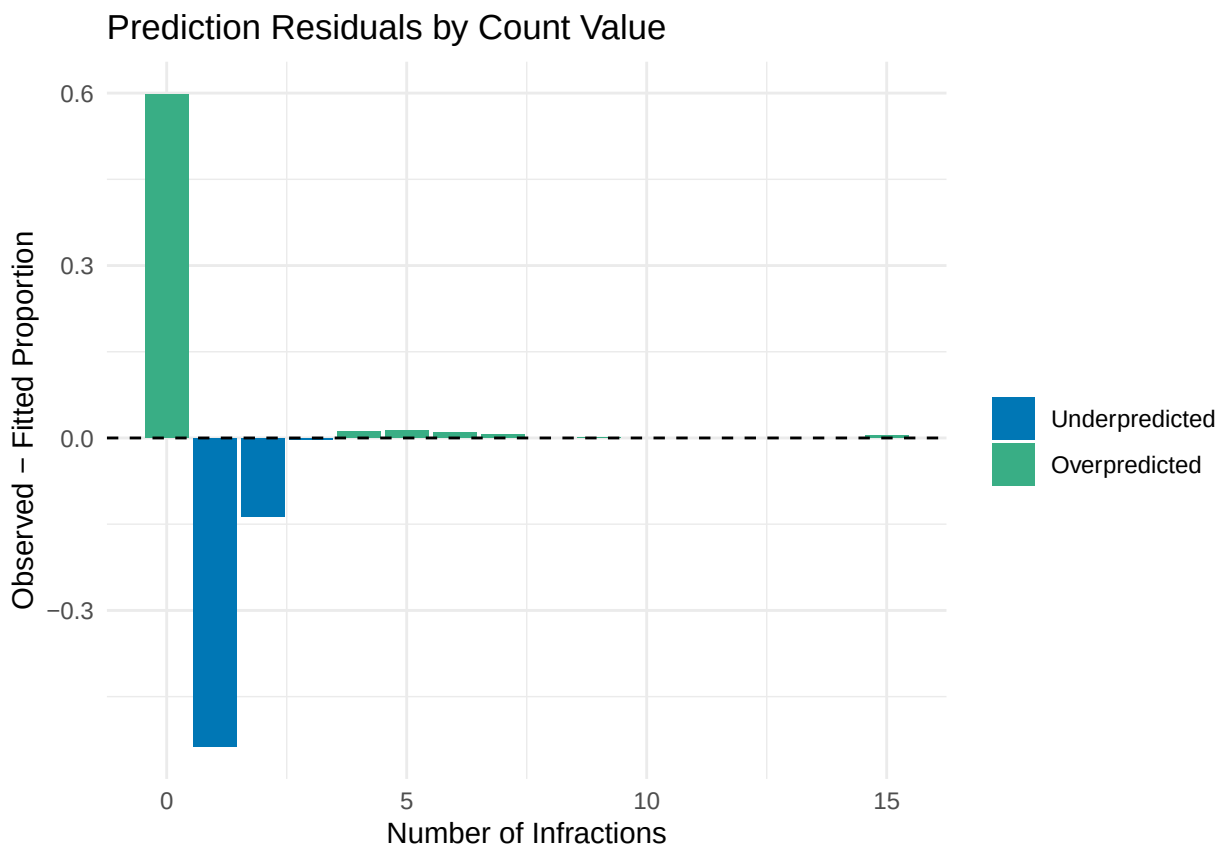
Table 3: Model Comparison

Model	AIC	Deviance	df	Overdispersion
Poisson	35484.49	2373.456	23	8.3143945
Negative Binomial	19761.59	385.641	23	0.2250685

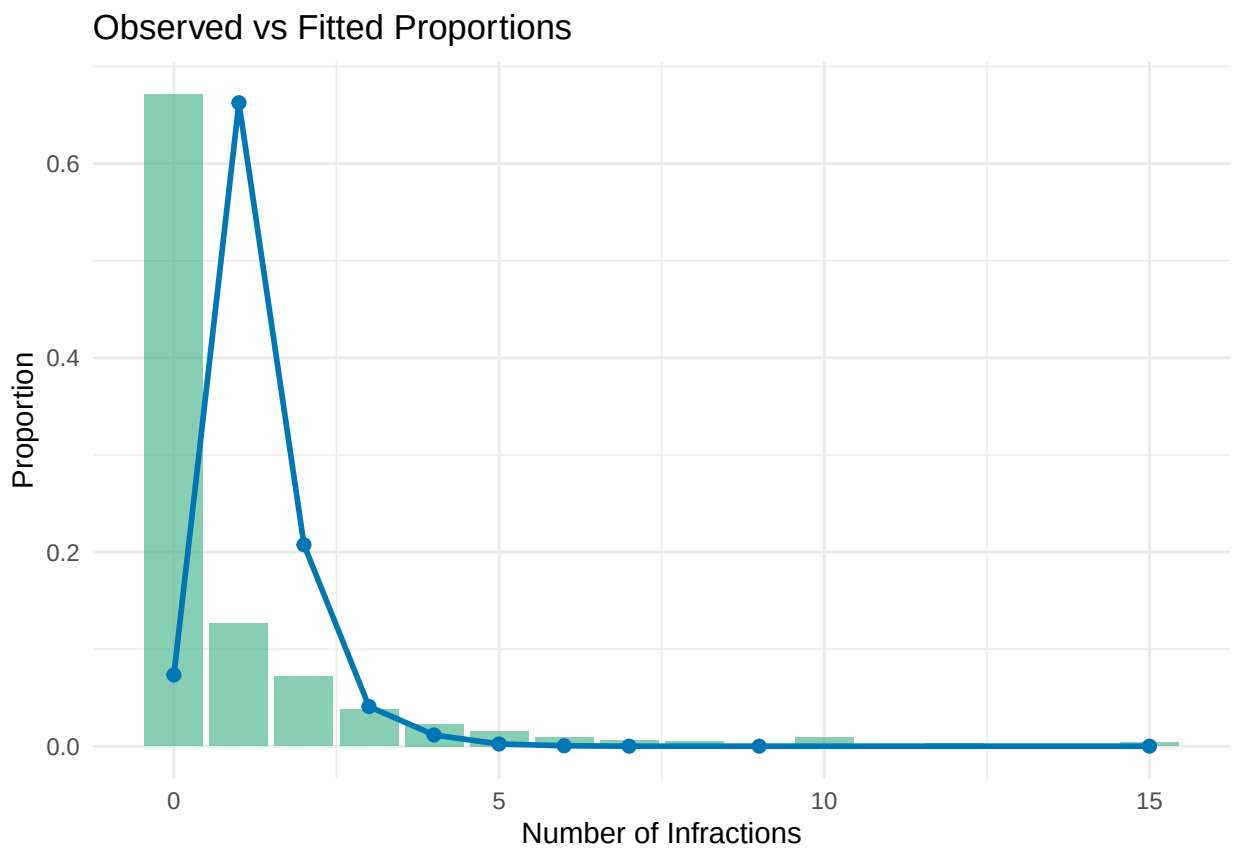
In our model, the most significant predictors are ADD/ADHD, Times Arrested, Children, Mental Health Treatment, Alcoholism, and an Unknown Offense Type. The variables with the largest effect size were ADD/ADHD with a 0.605 Coefficient and Unknown Offense type with -0.513 coefficient.

Table 4: Model Statistics

Variable	Estimate	Std.Error	P.Value	95% Confint IRR
ADD/ADHD	0.6047363	0.0637578	<0.0001	(1.616 - 2.074)
Times Arrested	0.0079125	0.0011492	<0.0001	(1.006 - 1.01)
Children	-0.3953537	0.0588810	<0.0001	(0.6 - 0.756)
Mental Health Treatment	0.3248295	0.0936386	0.00052	(1.152 - 1.663)
Alcoholism	0.2941803	0.0815311	0.00031	(1.144 - 1.575)
Offense Type: Unknown	-0.5126795	0.1334874	0.00012	(0.461 - 0.778)



Diagnostic plots suggest that the negative binomial model reproduces the broad distributional form of the outcome, but does not fully capture the allocation of probability among the smallest counts. The largest prediction errors occur at 0, 1, and 2 infractions, while fit improves for higher counts. This indicates that the model is adequate for the general shape of the count distribution but less accurate in modeling the concentration of low-count outcomes.



Interpretation

Our Stage One results indicate that high-risk classification in this sample is driven primarily by clinical factors and criminal justice history rather than socioeconomic variables. An ADD/ADHD diagnosis and receipt of post-admission mental health treatment are the strongest predictors of being labeled high risk. The coefficient for ADD/ADHD (≈ 1.34) corresponds to an odds ratio of about 3.8, meaning individuals with ADD/ADHD have nearly four times the odds of being classified as high risk compared to those without this diagnosis, holding other covariates constant. The coefficient for mental health treatment (≈ 2.02; odds ratio ≈ 7.5) indicates that inmates who receive mental health treatment post-admission have more than seven times the odds of being classified as high risk relative to those who do not receive treatment. Criminal justice variables also play an important role: each additional prior arrest is associated with a modest but statistically significant increase in the odds of high-risk classification (about 1–2% per arrest, cumulating over many arrests), and each additional child is associated with roughly a 9% increase in the odds of high-risk status. Property and public-order offenses exhibit higher odds of high-risk classification than the reference offense type, while years of education are not significantly related to risk status. Notably, the coefficient for male gender is negative and large in magnitude (odds ratio ≈ 0.39), implying that, conditional on diagnoses, treatment, offense type, and criminal history, males in our sample are less likely than the reference gender category to be classified as high risk. In Stage 2, we restricted the sample to inmates classified as "High Predicted Risk" and modeled the count of institutional infractions using a negative binomial regression. Within this high-risk subgroup, both structural disadvantage and clinical burden are central determinants of infraction counts. Each additional prior arrest and each additional mental health diagnosis are associated with approximately a 7% increase in the expected number of infractions (incidence-rate ratios ≈ 1.07), suggesting that deeper criminal justice involvement and greater psychiatric complexity are linked to more institutional misconduct. In contrast, years of education are protective once we condition on being high risk: each additional year of schooling is associated with about a 7% reduction in expected infractions (IRR ≈ 0.93),

implying that more educated high-risk inmates accumulate fewer infractions. Alcoholism is associated with substantially higher infraction counts (IRR ≈ 1.34), whereas ADD/ADHD is no longer a significant predictor once we restrict to the high-risk group. Demographically, male inmates have higher infraction counts than the reference gender group (IRR ≈ 1.5), and Black inmates exhibit markedly higher infraction counts than the reference race category (IRR on the order of 2.8), indicating large racial disparities in disciplinary outcomes among high-risk inmates even after adjustment for clinical and criminal-history covariates. Offense type is less influential in Stage 2; most categories are not significantly associated with infraction counts, with the exception of an "unknown" offense type that is associated with fewer infractions.

Treatment variables display different patterns. Mental health treatment remains positively associated with infraction counts among high-risk inmates. The coefficient of roughly 0.31 (IRR ≈ 1.37) indicates that treated high-risk inmates have about 37% more infractions than untreated high-risk inmates, conditional on diagnoses and other controls. An interaction between the number of diagnoses and mental health treatment is small and non-significant, suggesting that mental health treatment does not materially alter the relationship between diagnostic burden and infractions in our data. By contrast, alcohol/drug treatment shows a clinically meaningful interaction with alcoholism. While the main effect of alcohol/drug treatment is negligible overall, the Alcoholism $\times$ Alcohol/Drug Treatment interaction term is negative and statistically significant (IRR ≈ 0.76), indicating that among inmates with alcoholism, those who receive alcohol/drug treatment have approximately 24% fewer expected infractions than alcoholic inmates who do not receive such treatment.

Our first research question asked which variables best distinguish inmates with high versus low predicted infraction risk, with particular interest in demographic, criminal justice, and socioeconomic characteristics, and in whether inmates received mental health treatment post-admission. The Stage 1 logistic regression results provide a clear answer. High-risk classification is primarily shaped by clinical and justice-system variables—ADD/ADHD, alcoholism, receipt of mental health treatment, number of prior arrests, and offense type—rather than by socioeconomic status as proxied by years of education. Mental health treatment is strongly and positively associated with high-risk status, implying that mental health services are being targeted toward inmates whom the system already identifies as behaviorally or clinically high risk. Gender and family structure also contribute: males are less likely to be flagged as high risk once other factors are controlled, and inmates with more children are more likely to be classified as high risk. These findings address the first stage of our questions by demonstrating which factors the model uses to separate high- and low-risk inmates and how mental health treatment is distributed across those groups. Our second research question focused on the determinants of infraction severity within the high-risk group and the extent to which patterns in mental health and substance-use treatment reveal potential gaps in services. The Stage 2 negative binomial model speaks directly to this. Within the high-risk subgroup, higher infraction counts are associated with more prior arrests, more mental health diagnoses, alcoholism, lower levels of education, male gender, and Black race. Thus, our results identify a set of demographic and clinical factors—especially cumulative diagnostic burden and substance-use disorders—that drive higher infraction risk among already high-risk inmates. The treatment variables further inform our understanding of potential gaps. Mental health treatment is highly concentrated among inmates with greater clinical and behavioral severity and is positively associated with infractions, which is consistent with severity-driven selection into services rather than clear behavioral improvement. In contrast, the interaction between alcoholism and alcohol/drug treatment suggests a potentially beneficial effect of substance-use services among those with alcoholism, as treated alcoholic inmates have fewer infractions than untreated alcoholic inmates. These patterns address our second question by highlighting both who is most at risk within the high-risk population and where current mental health and substance-use services may or may not be reducing that risk.

Based on these models, we conclude that high-risk classification in this facility is closely tied to specific clinical conditions and criminal-history indicators. Inmates with ADD/ADHD, alcoholism, more arrests, more children, and certain offense types are substantially more likely to be flagged as high risk, whereas years of education do not play a meaningful role at the classification stage. Among those already classified as high risk, the intensity of institutional misconduct is shaped by cumulative mental health burden, substance-use status, criminal justice history, and structural disadvantage. In particular, additional diagnoses and prior arrests are associated with more infractions, alcoholism is a strong risk factor, and more years of education are associated with fewer infractions. The higher infraction rates observed among male and Black high-risk inmates point to important gender and racial disparities in disciplinary outcomes that warrant further

investigation.

With respect to services, our findings suggest that mental health treatment is being targeted toward the highest-risk inmates but, in these observational models, is not associated with lower infraction counts; the positive association likely reflects confounding by severity rather than a harmful effect of treatment. By contrast, alcohol/drug treatment appears to be behaviorally beneficial within the subgroup of inmates with alcoholism, as indicated by the significant interaction term. Taken together, these results support policy recommendations to intensify screening and treatment for alcohol use disorders among high-risk inmates, to expand or strengthen education-related programming for this population, to re-examine the design and delivery of mental health services with respect to behavioral outcomes, and to scrutinize racial and gender disparities in disciplinary practices and infraction recording.

Limitations

Conclusion

Appendix A: Data Dictionary

Table 5: Codebook: Variable Descriptions and Coding

Variable Name	Original Code	Type	Description	Coding	Source
Derived Variables					
at_risk	Derived	Binary (0/1)	Binary indicator from median split of logistic risk probability	1 = Above median predicted risk probability, 0 = Below median	Derived from Stage 1 logistic model predictions
num_diagnoses	Derived	Numeric	Count of 6 specific mental health diagnoses	0-6 (sum of: Bipolar, Depression, Schizophrenia, PTSD, Anxiety, and Personality Disorder)	Derived from psychiatric screening
MHTx:num_diagnoses	Interaction	Numeric	Mental Health Treatment x Diagnosis Count interaction	Product: Mental Health Treatment x num_diagnoses	Model-derived interaction
Alcoholism:DrugTx	Interaction	Numeric	Alcoholism x Substance Treatment interaction	Product: Alcoholism x Alcohol/Drug Treatment	Model-derived interaction
Outcome & Continuous Predictors					
Number Infractions	V1398	Numeric	Total count of disciplinary infractions during incarceration	Continuous (0-250+)	Prison disciplinary records
Times Arrested	V0909	Numeric	Lifetime number of prior arrests	Continuous (count, negatives → NA)	Criminal history records
Years of Education	V0935	Numeric	Total years of formal education completed	Continuous (years, negatives → NA)	Self-reported education history
Categorical Predictors					
Race	V1951-V1957	Factor	Self-reported race/ethnicity at intake	White/Black/Hispanic/Other (Multiracial/Asian/Native Hawaiian/AI/Other collapsed)	Demographic intake form
Gender	V1212	Factor	Self-described gender identity	Male/Female/Transgender/Do not identify as male, female or transgender	Demographic intake form
offense_type	V0062	Factor	Primary instant offense category	Violent/Property/Drug/Public Order	Presentence investigation report
Marital Status	V0022	Factor	Marital status at time of offense	0-5 (coded levels: Single/Married/Divorced/etc.)	Family status questionnaire
Binary Predictors					
ADD/ADHD	V0942	Binary (0/1)	Lifetime diagnosis of ADD/ADHD	0 = No/Never, 1 = Yes (2 → 0 recoded)	Psychiatric screening questionnaire
Bipolar	V1185	Binary (0/1)	Lifetime bipolar disorder diagnosis	0 = No, 1 = Yes (2 → 0 recoded)	Psychiatric screening
Depression	V1186	Binary (0/1)	Lifetime major depression diagnosis	0 = No, 1 = Yes (2 → 0 recoded)	Psychiatric screening
Schizophrenia	V1187	Binary (0/1)	Lifetime schizophrenia diagnosis	0 = No, 1 = Yes (2 → 0 recoded)	Psychiatric screening
PTSD	V1188	Binary (0/1)	Lifetime PTSD diagnosis	0 = No, 1 = Yes (2 → 0 recoded)	Psychiatric screening
Anxiety	V1189	Binary (0/1)	Lifetime anxiety disorder diagnosis	0 = No, 1 = Yes (2 → 0 recoded)	Psychiatric screening
Personality Disorder	V1190	Binary (0/1)	Lifetime personality disorder diagnosis	0 = No, 1 = Yes (2 → 0 recoded)	Psychiatric screening
Other Diagnosis	V1191	Binary (0/1)	Other lifetime mental health diagnosis	0 = No, 1 = Yes (2 → 0 recoded)	Psychiatric screening
Alcoholism	V1265	Binary (0/1)	Lifetime alcohol dependence diagnosis	0 = No, 1 = Yes (2 → 0 recoded)	Substance use disorder assessment
Armed Forces	V0023	Binary (0/1)	U.S. military service history	0 = No, 1 = Yes (2 → 0 recoded)	Military service verification
Homeless	V0958	Binary (0/1)	Housing instability prior to arrest	0 = Stably housed, 1 = Homeless (2 → 0 recoded)	Pretrial housing assessment
Children	V0982	Binary (0/1)	Has minor dependent children	0 = No, 1 = Yes (2 → 0 recoded)	Family responsibility assessment
Mental Health Treatment	V1204	Binary (0/1)	Received mental health treatment during incarceration	0 = No treatment received post-admission, 1 = Received MHT during prison stay (2 → 0 recoded)	Prison mental health treatment records

Table 5: Codebook: Variable Descriptions and Coding *(continued)*

Variable Name	Original Code	Type	Description	Coding	Source
Alcohol/Drug Treatment	V1374	Binary (0/1)	Received substance abuse treatment during incarceration	0 = No treatment received post-admission, 1 = Received substance treatment during prison (2 → 0 recoded)	Prison substance abuse treatment records

Appendix B: Model Summary Outputs

```
## Stage 1 Logistic Model

##
## Call:
## glm(formula = at_risk ~ Years.of.Education + ADD.ADHD + Gender +
##     offense_type + Children + Times.Arrested + Mental.Health.Treatment +
##     Alcoholism, family = binomial(link = "logit"), data = cleaned2,
##     na.action = na.exclude)
##
## Coefficients:
##                           Estimate Std. Error z value Pr(>|z|)
## (Intercept)              -0.105439   0.111897  -0.942 0.346044
## Years.of.Education       -0.006716   0.007237  -0.928 0.353398
## ADD.ADHD                  1.338996   0.047447  28.221  < 2e-16 ***
## GenderMale               -0.938654   0.047758 -19.654  < 2e-16 ***
## offense_typeOther         0.260657   0.350827   0.743 0.457494
## offense_typeProperty      0.235999   0.067435   3.500 0.000466 ***
## offense_typePublic Order  0.193383   0.074804   2.585 0.009732 **
## offense_typeUnknown       0.217762   0.090668   2.402 0.016317 *
## offense_typeViolent       0.120553   0.056052   2.151 0.031498 *
## Children                 -0.086372   0.042271  -2.043 0.041025 *
## Times.Arrested            0.013053   0.001667   7.829 4.94e-15 ***
## Mental.Health.Treatment   2.201666   0.050328  43.746  < 2e-16 ***
## Alcoholism                0.087787   0.038603   2.274 0.022961 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 21322  on 15449  degrees of freedom
## Residual deviance: 16637  on 15437  degrees of freedom
##   (1708 observations deleted due to missingness)
## AIC: 16663
##
## Number of Fisher Scoring iterations: 4

## Stage 2 Logistic Model

##
## Call:
## glm.nb(formula = Number.Infractions ~ Race + num_diagnoses +
##     Armed.Forces + +Times.Arrested + Years.of.Education + ADD.ADHD +
##     Homeless + Children + Mental.Health.Treatment + Mental.Health.Treatment:num_diagnoses +
##     Alcoholism + Alcohol.Drug.Treatment + Alcoholism:Alcohol.Drug.Treatment +
##     Gender + offense_type, data = high_risk_group, init.theta = 0.225068483,
##     link = log)
##
## Coefficients:
##                                      Estimate Std. Error z value Pr(>|z|)
## (Intercept)                         -0.656952   0.519424  -1.265 0.205953
## RaceBlack                            1.035531   0.490403   2.112 0.034721 *
```

```
## RaceHispanic                          1.054957   0.511488   2.063 0.039158 *
## RaceMultiracial                       0.730125   0.489379   1.492 0.135714
## RaceOther                             1.330617   0.536754   2.479 0.013175 *
## RaceWhite                             0.439531   0.489300   0.898 0.369034
## num_diagnoses                         0.072018   0.026254   2.743 0.006086 **
## Armed.Forces                         -0.145850   0.119248  -1.223 0.221300
## Times.Arrested                        0.007912   0.001149   6.885 5.76e-12 ***
## Years.of.Education                   -0.037718   0.011595  -3.253 0.001142 **
## ADD.ADHD                              0.604736   0.063758   9.485  < 2e-16 ***
## Homeless                             -0.114640   0.113859  -1.007 0.314003
## Children                             -0.395354   0.058881  -6.714 1.89e-11 ***
## Mental.Health.Treatment               0.324830   0.093639   3.469 0.000522 ***
## Alcoholism                            0.294180   0.081531   3.608 0.000308 ***
## Alcohol.Drug.Treatment                0.023718   0.074325   0.319 0.749638
## GenderMale                           -0.045294   0.062580  -0.724 0.469206
## offense_typeOther                     0.400541   0.417414   0.960 0.337268
## offense_typeProperty                 -0.014625   0.096488  -0.152 0.879527
## offense_typePublic Order             -0.240554   0.111770  -2.152 0.031380 *
## offense_typeUnknown                  -0.512680   0.133487  -3.841 0.000123 ***
## offense_typeViolent                   0.125330   0.085486   1.466 0.142624
## num_diagnoses:Mental.Health.Treatment -0.014916   0.033334  -0.447 0.654547
## Alcoholism:Alcohol.Drug.Treatment     -0.281589   0.109103  -2.581 0.009853 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(0.2251) family taken to be 1)
##
##     Null deviance: 5716.9  on 7618  degrees of freedom
## Residual deviance: 5331.2  on 7595  degrees of freedom
##   (105 observations deleted due to missingness)
## AIC: 19762
##
## Number of Fisher Scoring iterations: 1
##
##
##               Theta:  0.22507
##           Std. Err.:  0.00655
##
##  2 x log-likelihood:  -19711.59400
```